True error = E_t = true val - approx

$\varepsilon_t = \frac{\text{true error}}{\text{true value}}\%$ } true percent relative error

$\varepsilon_a = \frac{\text{approx. error}}{\text{approx.}}\%$ $|\varepsilon_a| < \varepsilon_s$

$\varepsilon_s = (0.5) \times 10^{2-n}\%$

$(-1)^s \times (1.f)_2 \times 2^{exp-127}$

IEEE 32bit: 24b mantissa

7 significant digits $10^{-38} - 10^{39}$

64 bit: 15-16 decimal precision

$10^{-308} - 10^{308}$ $m \cdot b^e$

Truncation err: when approx used.

Roundoff err: limited signi. figures

Catastrophic cancel.: substracting nearly equal numbers

$1/b \le m < 1$ $\varepsilon_m = b^{1-t}$

$1 + \delta = 1$ whenever $|\delta| < \varepsilon_m$

Taylor's th.:

$f(x) = f(a) + f'(a)(x-a) + \frac{1}{2!} f''(2a)(x-a)^2 + \frac{1}{3!} f'''(a)(x-a)^3 + \frac{1}{n!} f^{(n)}(a)(x-n)^n + R_n$

$R_n = \int_a^x \frac{(x-t)^n}{n!} f^{(n+1)}(t)\,dt$ t is dummy

$f(x_{i+1}) \cong f(x_i)$: zero order approx (const+1)

$f(x_{i+1}) \cong f(x_i) + f'(x_i)(x_{i+1} - x_i)$: 1st or. (straight line 1st order approx)

$(x_{i+1} - x_i) = h$ = step size

2nd order approximation:

$f(x_{i+1}) \cong f(x_i) + f'(x_i)h + f''(x_i)(h^2/2!)$

$h = 0.5$ $x_{i-1} = 0$ $x_i = 0.5$ $x_{i+1} = 1.0$

$f(x_{i-1}) = 1.25 = f(0)$

$f(x_i) = 0.925 = f(0.5)$

$f(x_{i+1}) = f(1)$ Forward diff:

$f(0.5) \cong \frac{f(1) - f(0.5)}{1 - 0.5}$ Centered:

$f'(0.5) \cong \frac{f(1) - f(0)}{1 - 0}$

$\tilde{x} = 2.5$ $\Delta\tilde{x} = 0.01$ $f(x) = x^3$

est. resulting error

$\Delta f(\tilde{x}) = |f'(\tilde{x})|\,\Delta\tilde{x}$

$= 3\tilde{x}^2 \Delta\tilde{x}\,|_{\tilde{x}=2.5}$

Relative error $\frac{x - \tilde{x}}{x}$

Condition num.: $\frac{\tilde{x} f'(\tilde{x})}{f(\tilde{x})}$

cond num = 1 func. rel. err. is identical to rel. err of x

cond num > 1 rel err. is amplified

Bisection Method

$f(x_e) f(x_u) < 0$

$x_r = \frac{x_e + x_u}{2}$

$\varepsilon_a = \left| \frac{x_r^{new} - x_r^{old}}{x_r^{new}} \right| \%$

$\varepsilon_a < eps$ stop

False-Position

$x_r = x_u - \frac{f(x_u)(x_e - x_u)}{f(x_e) - f(x_u)}$

Fixed Point

$x = g(x)$ $x_{i+1} = g(x_i)$

$e^{-x} - x = 0 \Rightarrow y_1 = x$ $y_2 = e^{-x}$

Converges if $|g'(x)| < 1$ around the root

Newton-Raphson

$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$

$E_{t,i+1} = -\frac{f''(x_r)}{2f'(x_r)} E_{t,i}^2$

x_r true root

Secant Method

$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$

- Modified:

δ: a small perturbation fraction

$x_{i+1} = x_i - \frac{\delta x_0 \cdot f(x_0)}{f(x_i + \delta x_i) - f(x_i)}$

Brent's Method

Hybrid method combine reliability of bracketing and speed of open

(x_{i-2}, y_{i-2}), (x_{i-1}, y_{i-1}), (x_i, y_i)

$g(y) = \frac{(y-y_{i-1})(y-y_i)}{(y_{i-2}-y_{i-1})(y_{i-2}-y_i)} x_{i-2} + \frac{(y-y_{i-2})(y-y_i)}{(y_{i-1}-y_{i-2})(y_{i-1}-y_i)} x_{i-1}$

$+ \frac{(y-y_{i-2})(y-y_{i-1})}{(y_i-y_{i-2})(y_i-y_{i-2})} x_i$ the root corresponds to $y = 0$:

$x_{i+1} = \frac{y_{i-1} y_i}{(y_{i-2}-y_{i-1})(y_{i-2}-y_i)} x_{i-2} + \frac{y_{i-2} \cdot y_i}{(y_{i-1}-y_{i-2})(y_{i-1}-y_i)} x_{i-1} + \frac{y_{i-2}\, y_{i-1}}{(y_i-y_{i-2})(y_i-y_{i-2})} x_i$

False Position

$x_r = \frac{af(b) - bf(a)}{f(b) - f(a)}$

$x = \frac{2c}{-b \mp \sqrt{b^2 - 4ac}}$

Finite Difference Approx.

Taylor series expansion for $f(x_{i+2})$

$f(x_{i+2}) = f(x_i) + f'(x_i)(2h) + \frac{f''(x_i)}{2!}(2h)^2 + \dots$ (2)

Multiply (1) by 2 add subtract from (2)

$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2} + O(h)$ } 2nd forward finite difference

$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2})}{h^2} + O(h)$ } backward

$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2} + O(h)$ } centered

MATLAB

```
x = fzero(@(x) x^2 - 9, -4) % finds negative root
c = [.....] % coefficients of polynomial polyval(c, 1) %x=1
x = roots(c)  a = poly(x)
x0 = ... % initial guess; k=0; xnew = x0;
while ( abs( (xnew - xold)/xold ) ) > delta ) & k < maxit
  xold = xnew;
  xnew = cos(xold);
  k = k+1;
end
```

Absolute convergence criterion:

$|x - x_{old}| < \Delta_a$ Δ_a: tolerance

Relative: $|(x - x_{old})/x_{old}|$

$eps = 2.2204 \times 10^{-16}$

$\frac{d}{dx}\tan x = \sec^2 x$ $\sec x = \sec x \tan x$ $\csc x = -\csc x \tan x$ $\cot x = -\csc^2 x$

$\sin^{-1} x = 1/\sqrt{1-x^2}$ $\cos^{-1} x = -1/\sqrt{1-x^2}$ $\tan^{-1} x = 1/x^2 + 1$

$a^x = a^x \ln a$ $e^x = e^x$ $\ln x = 1/x$ $\log_a x = 1/x \ln a$

There is no general convergence criteria for Newton-Raphson. Converge depends on nature of function, accuracy of initial guess

Fixed point major weakness: one-side. one of the bracketing pts stay fixed

$\varepsilon_a > \varepsilon_t$ always for bisection method